Niranjan Khedkar

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Summary

AI Engineer with 3 years at IBM building and operating production GenAI systems for global enterprise clients. Hands-on across the full lifecycle: LLM fine-tuning, RAG and agentic pipelines, GPU inference optimization, containerized deployment on Kubernetes/OpenShift, and multi-cloud production operations. Delivered 15+ production AI systems generating $10M+ in revenue for Japanese and Fortune 500 clients.

Skills

Languages: Python, SQL, Golang, JavaScript, Bash

GenAI & LLMs: LLM Fine-tuning (LoRA, SFT), RAG, Agentic AI (CrewAI, LangGraph), LangChain, LlamaIndex, MCP, vLLM, Hugging Face, Prompt Engineering, Vector Databases (Milvus, ChromaDB, FAISS), LLM Evaluation (RAGAS, BLEU, COMET, BLEURT)

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers (BERT), YOLO (Object Detection), OpenCV, Multi-modal AI (Image-to-Text)

MLOps & Cloud: Docker, Kubernetes, OpenShift, Helm, AWS, GCP, Azure, CI/CD, Monitoring & Inference Optimization

Data & Backend: Pandas, NumPy, PostgreSQL, FastAPI, Flask, Django, REST APIs, Streamlit

Spoken Languages: English (Native), Japanese (JLPT N4, working proficiency with clients)

Experience

IBM — Bengaluru, India

AI Operations Delivery Consultant, Expert Labs | OpenShift, Kubernetes, Python, AWS, GCP, Azure — *Nov 2025 – Present*

- Architect and operate production AI/ML workloads on OpenShift and Kubernetes, owning orchestration, autoscaling, and resource optimization for containerized inference services.
- Operate enterprise AI systems for Fortune 500 clients including **UPS, Adobe, and AT&T**, maintaining **99.9%+ uptime** for mission-critical automation workloads.
- Fine-tune deployed models and automation workflows against production feedback, improving inference speed, accuracy, and resource consumption per client SLAs.
- Troubleshoot complex production incidents across multi-cloud environments (AWS, GCP, Azure, IBM Cloud), reducing mean time to resolution through systematic debugging and runbook frameworks.

IBM — Bengaluru, India

AI Engineer | Python, LangGraph, CrewAI, RAG, Docker, OpenShift — *Oct 2023 – Oct 2025*

- Architected and deployed **15+ production GenAI systems worth $10M+ in revenue** for Japanese enterprise clients (Toyota, Honda, and financial institutions), using enterprise LLMs with RAG pipelines and autonomous agents.
- Reduced manual workflows by **40%** by orchestrating multi-agent systems with CrewAI and LangGraph that automated research, code generation, and business processes at enterprise scale.
- Built hybrid-cloud AI infrastructure with Docker and OpenShift, standardizing containerized deployment of LLM inference and RAG services across client environments.
- Delivered end-to-end MLOps pipelines covering data ingestion, fine-tuning, evaluation, deployment, and monitoring.
- Delivered GenAI applications spanning AI-assisted code generation and WhatsApp conversational bots for client-facing business automation.
- Built computer vision and multi-modal AI solutions, including image-to-text generation pipelines and YOLO-based symbol detection and classification for nuclear plant engineering diagrams.

Richin AI — Mumbai, India (Remote)

AI Engineer (Freelance) | Python, NLP, Chatbots — *Jul 2023 – Aug 2023*

- Designed and deployed a customer-facing chatbot for a finance company, cutting response time by **40%** and improving sentiment analysis accuracy by **18%**.

Projects

Straker AI Translation Platform – ESA Partner | *vLLM, NVIDIA A100, Gemma 2/3, LoRA, SFT* — May 2025 – Nov 2025

- Primary technical resource for Straker's neural translation platform; migrated the serving stack from AWS to cloud GPU infrastructure (2× NVIDIA A100 80GB) via a BYOM vLLM runtime with continuous batching.
- Designed an asynchronous proxy layer that load-balanced and dispatched batch translation requests in parallel across two vLLM model replicas, achieving **6.2× inference throughput** (2712s → 280s for 2,000 segments).
- Built an end-to-end fine-tuning pipeline with Hugging Face: corpus prep, base model benchmarking (Gemma 2 9B / Gemma 3 12B/27B), LoRA + SFT continual pre-training, and multi-metric evaluation (BLEU, COMET, BLEURT, MetricX-24 XL); delivery across three time zones **secured the ESA deal**.

Project Hail Mary – Offline Audio Translator | *Python, Librosa, FAISS, ONNX Runtime, Streamlit* — Personal

- Built a fully offline sound-to-word translator inspired by Andy Weir's novel: extracts 40 MFCC features per audio clip via Librosa, indexes them in FAISS, and retrieves the closest match by L2 distance.

Education

Pillai College of Engineering, University of Mumbai — Mumbai, India

Bachelor of Technology (CGPA: 9.61/10) — *Jul 2019 – Jul 2023*

Publications, Achievements & Certifications

Publication: *Determination of Medicinal Leaf Properties*, IJSREM – CNN-based plant classification research (94% accuracy)

IBM Employee Equity Program (EEP) Award | **IBM 100% Club Member** | **JLPT N4 Certified** (Japanese)